UNIT-V

The Transport Layer & Application Layer

The Transport Layer: Transport layer protocols: Introduction – services – port number – User Datagram Protocol (UDP) – User Datagram Protocol (UDP) services – UDP applications – Transmission Control Protocol (TCP): TCP services – TCP features – Segment – A TCP connection – Windows in TCP – Flow control – Error control – Congestion control in TCP.

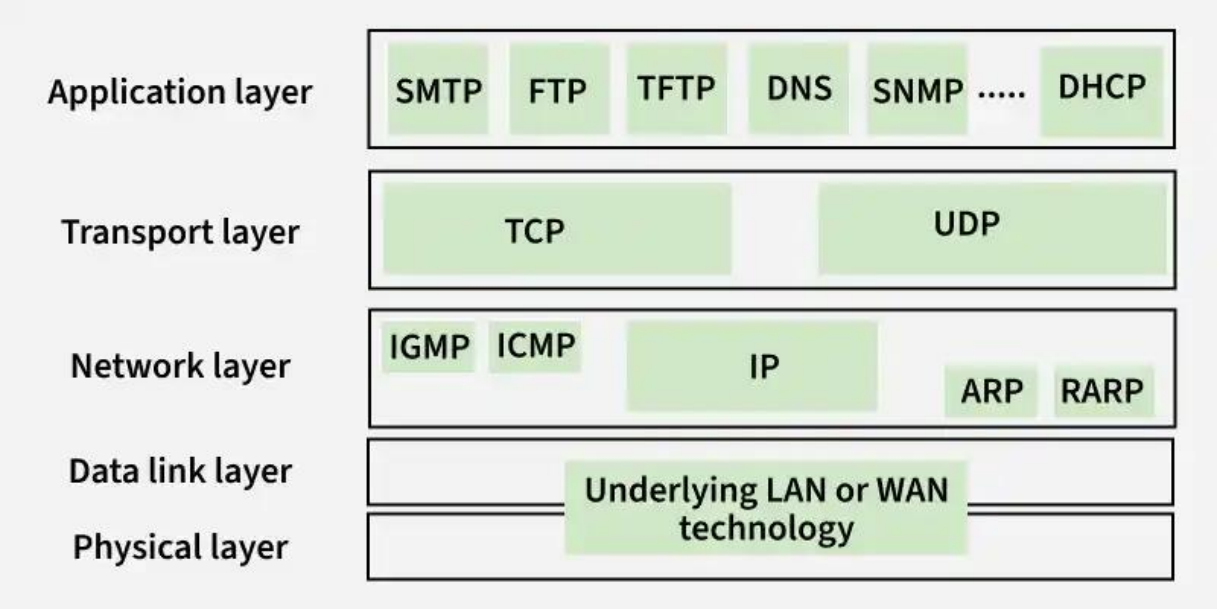
Application Layer: World Wide Web: HTTP, Electronic mail – Architecture – Web-based mail – Email security – TELNET – Local versus remote logging – Domain Name System (DNS).

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Transport layer protocols:

5.1 Introduction:

Transport layer protocols are communication protocols that enable end-to-end data transfer between applications on different devices.Operate between the application layer and network layer. Enable process-to-process communication using port numbers Determine how data is transmitted, whether through reliable or fast delivery methods Provide reliability, flow control, and error handling when required.



Functions of Transport Layer

- **Process-to-Process Delivery:** Ensures data is delivered between the correct sender and receiver processes.
- **End-to-End Connection:** Establishes and maintains a logical connection between communicating devices.

- **Multiplexing and Demultiplexing:** Allows multiple applications to send and receive data simultaneously using port numbers.
- **Data Integrity and Error Correction:** Detects and corrects errors to ensure reliable data transmission.
- **Congestion Control:** Controls data flow to prevent network congestion.

5.2 Characteristics of Transport Layer Protocol

- The transport layer uses two main protocols: TCP and UDP.
- It enables process-to-process communication between source and destination hosts.
- Data from the network layer is delivered to the correct application using port numbers.
- Each port is identified by a 16-bit port number, allowing multiple applications to run simultaneously.
- TCP provides reliable, connection-oriented communication with error control and flow control.
- UDP provides fast, connectionless communication with minimal overhead

5.3 Transport Layer Protocols

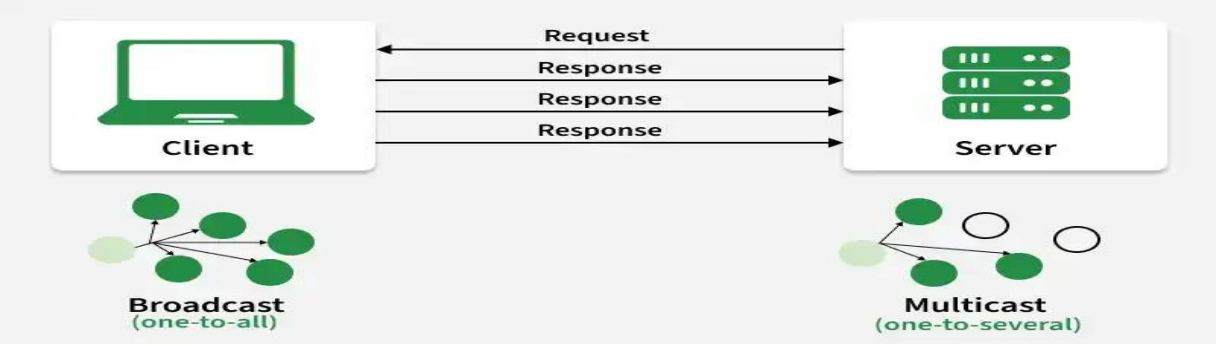
The transport layer is represented majorly by TCP and UDP protocols. Transport layer protocols work above the IP protocols and deliver the data packets from IP serves to destination port and from the originating port to destination IP services.

5.3.1 UDP

UDP (User Datagram Protocol) is a connectionless transport layer protocol that provides fast but unreliable data transmission. It is mainly used where speed is more important than reliability.

- **Features of UDP**

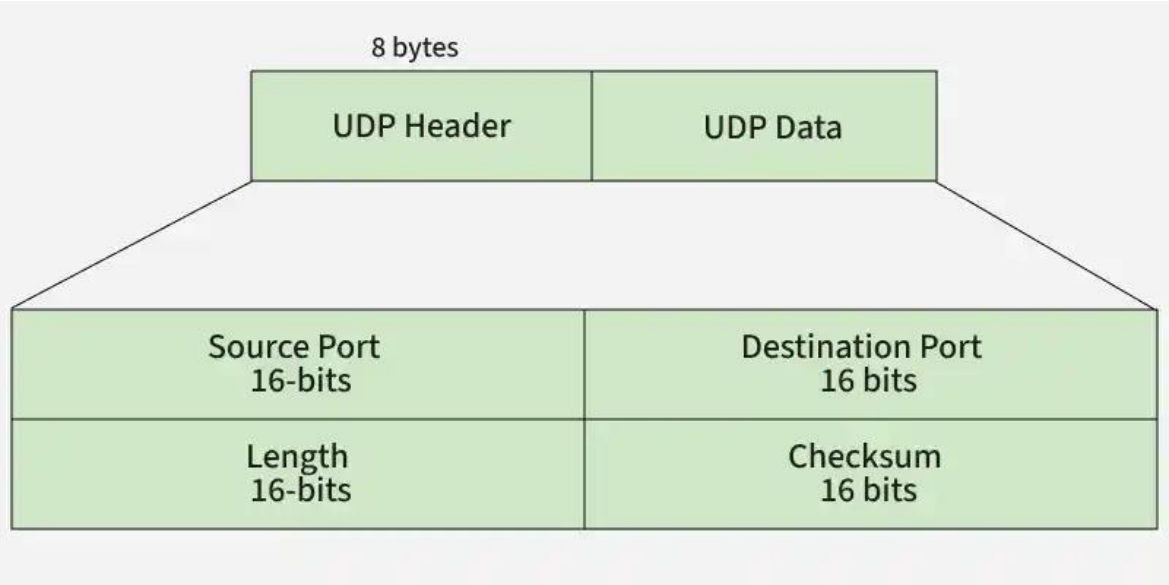
- Connectionless and non-sequential data transmission
- No acknowledgment or retransmission
- Low overhead and fast delivery
- Used in applications like DNS, streaming, and online games



- **UDP Segment**

Field	**Description**
Source Port	Identifies sender's port (16 bits)
Destination Port	Identifies receiver's port (16 bits)
Length	Total length of UDP header + data
Checksum	Error detection using one's complement

- **Port range:** 0 to 65535 (port 0 is reserved)



- **Advantages of UDP**

- Supports broadcast and multicast
- Faster data transmission
- Suitable for small transactions
- No need for connection setup

- **Disadvantages of UDP**

- No guarantee of data delivery
- No congestion control
- No sequencing of data
- Higher packet loss possible

➢ **UDP Services**

UDP provides the following services:

1. Process-to-process communication: UDP delivers data from a source application to the destination application using port numbers.

2. Connectionless service: UDP does not establish a connection or perform a handshake before sending data.

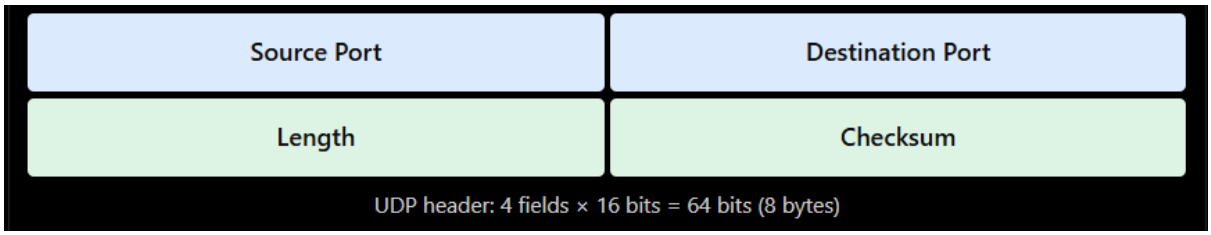
3. Multiplexing and demultiplexing: Port numbers allow multiple applications to send and receive data through the network simultaneously.

4. Error detection: UDP uses a checksum to detect errors in the UDP header and data. If corruption is detected, the datagram is normally discarded by the receiving system.

5. Message-oriented communication: UDP preserves message boundaries. Each UDP datagram is treated as a separate message.

6. Low overhead and fast transmission: UDP has a small header of only 8 bytes and does not provide acknowledgements, retransmissions, or flow and congestion control.

➢ **UDP header format:**



UDP Applications

UDP is commonly used in applications where low delay and fast data delivery are important.

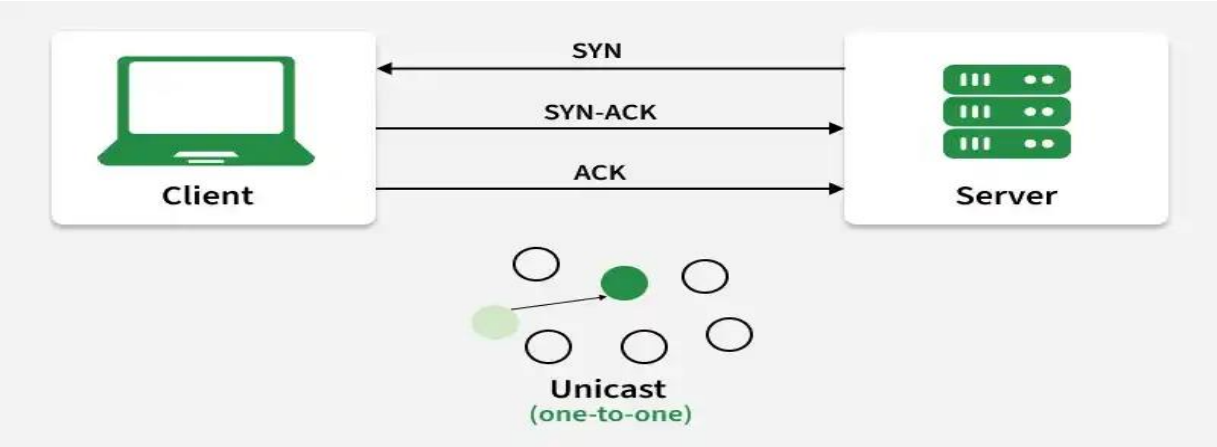
Application	Purpose of UDP
DNS (Domain Name System)	Sends queries to translate domain names into IP addresses.
DHCP	Assigns IP addresses and network configuration to devices.
VoIP	Transmits voice packets with low delay.
Live video streaming	Supports real-time audio and video delivery.

Online gaming	Sends frequent updates about player positions and game events.
TFTP	Provides simple file transfer using UDP.
NTP	Synchronizes clocks across networked devices.

Key point for examinations: UDP is a connectionless, message-oriented transport protocol that provides process-to-process delivery using port numbers, with minimal overhead but no guarantee of reliable delivery.

5.3.2 TCP:

TCP (Transmission Control Protocol.) is a connection-oriented and reliable transport layer protocol. It establishes a virtual connection before data transmission and ensures error-free delivery.



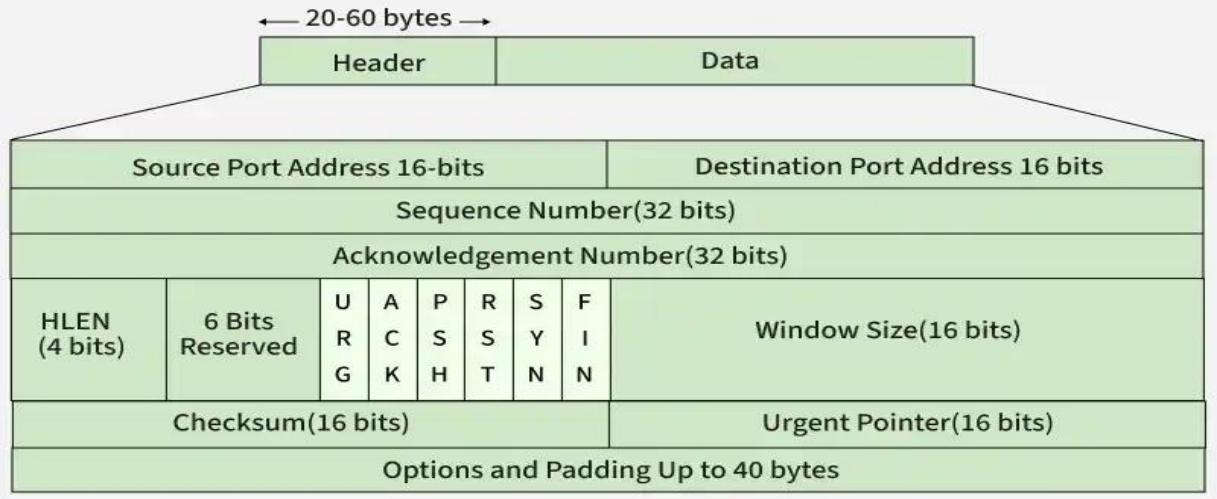
- **Features of TCP**

- Reliable and ordered data delivery
- Uses acknowledgments and retransmissions
- Data sent as a byte stream
- Uses sequence numbers for tracking data

- **TCP SEGMENT**

Field	Description
Source Port	Sender's application port
Destination Port	Receiver's application port
Sequence Number	Byte number of first data byte
Acknowledgement Number	Next expected byte

Field	Description
Header Length	Size of TCP header
Control Flags	URG, ACK, PSH, RST, SYN, FIN
Window Size	Flow control
Checksum	Error detection
Urgent Pointer	Urgent data indicator



- **Advantages of TCP**

- Reliable data transmission
- Error and flow control
- Connection-oriented communication
- Independent of operating system

- **Disadvantages of TCP**

- Slower compared to UDP
- No support for broadcast or multicast
- Higher overhead
- Difficult protocol replacement in TCP/IP model

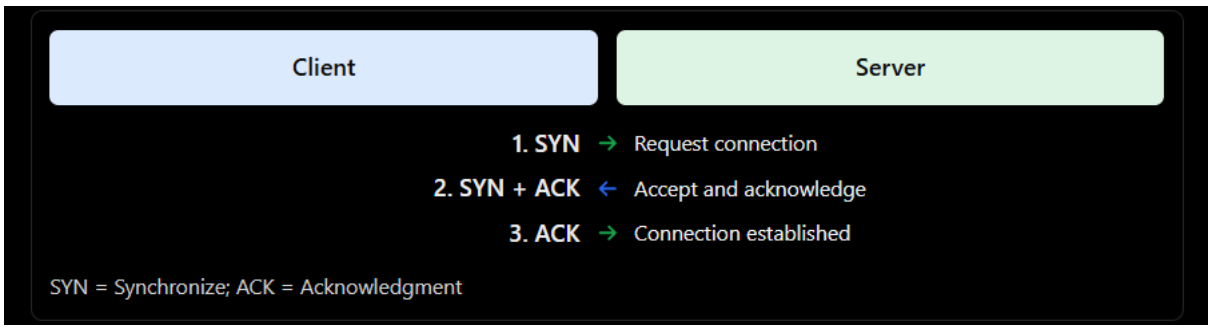
- **A TCP Connection:**

TCP (Transmission Control Protocol) is a connection-oriented transport layer protocol that provides reliable, ordered delivery of data between two devices.

Before transferring data, TCP establishes a connection between the sender and receiver. After communication is completed, the connection is terminated.

A. TCP connection establishment — Three-way handshake

TCP uses a three-way handshake to establish a connection.



- **Steps:**

1. SYN: The client sends a SYN segment to request a connection.
2. SYN-ACK: The server responds with a SYN-ACK segment.
3. ACK: The client sends an ACK segment, completing the handshake.

- **Data transfer**

Once the connection is established, data is transmitted in TCP segments. The receiver acknowledges received data, and TCP uses sequence numbers to maintain the correct order.

- **TCP connection termination**

TCP normally uses a four-segment exchange to close a connection gracefully:

1. One side sends FIN.
2. The other side acknowledges with ACK.
3. The other side sends FIN when ready.
4. The first side sends ACK.

FIN means Finish. TCP may also terminate a connection using a reset (RST) when necessary.

Windows in TCP

TCP uses a sliding window mechanism to control the amount of data that can be sent before acknowledgments are received.

The sender can transmit multiple bytes of data without waiting for an acknowledgment for every individual byte.

Types of TCP windows

1. Send window: Represents the range of sequence-numbered bytes that the sender is currently permitted to send.

2. Receive window: Indicates how much additional data the receiver can accept. It is advertised by the receiver and is used for flow control.

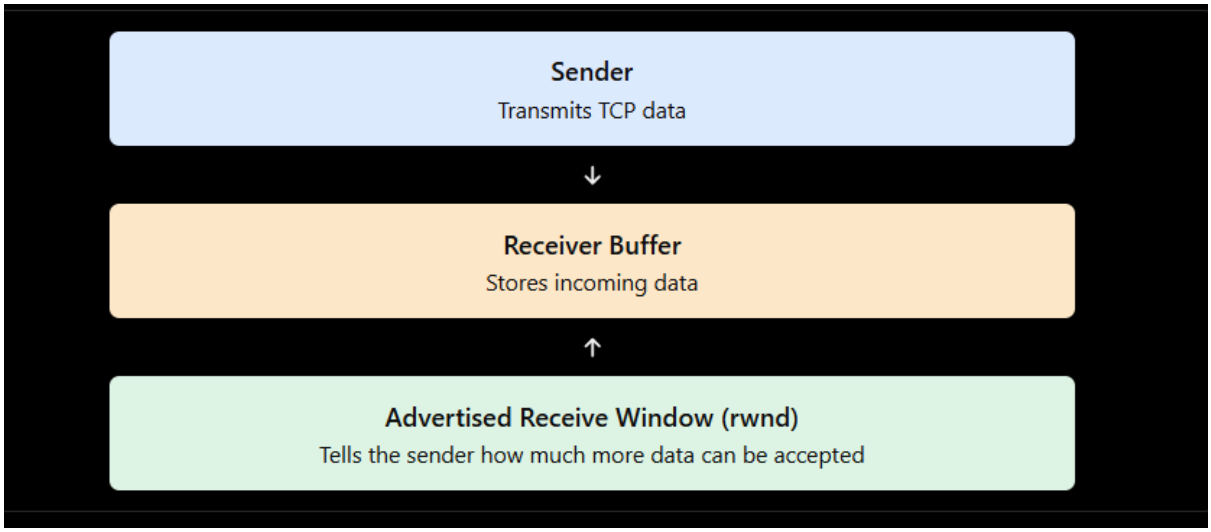
3. Sliding window: As acknowledgments arrive and buffer space becomes available, the permitted sending range advances, allowing more data to be transmitted.

Flow Control in TCP

Flow control prevents a fast sender from overwhelming a slow receiver.

The receiver has a limited buffer to store incoming data. If the sender transmits too much data too quickly, the receiver's buffer may become full.

How TCP flow control works



Working:

1. The receiver informs the sender of its available buffer space through the advertised receive window.
2. The sender limits outstanding data according to the receive window.
3. If the receiver has more buffer space, it advertises a larger window.
4. If the receiver's buffer becomes full, it can advertise a zero window, temporarily stopping new data transmission.

Example: If a receiver can accept only 2 KB of additional data, the sender must respect that limit even if it can transmit much faster.

Key point: Flow control protects the receiver from being overwhelmed.

Error Control in TCP

Error control ensures reliable and ordered delivery of data despite packet loss, corruption, duplication, or out-of-order arrival.

TCP uses the following mechanisms:

1. Checksum: Detects corruption in the TCP header and data.
2. Sequence numbers: Identify byte positions and help TCP arrange data in the correct order.

3. Acknowledgments (ACKs): Inform the sender about received data.
4. Retransmission: Lost or unacknowledged data is retransmitted when TCP determines it is necessary.
5. Timeout mechanism: If an acknowledgment does not arrive within the retransmission timeout, TCP retransmits the relevant data.
6. Duplicate ACKs and fast retransmit: Multiple duplicate acknowledgments can indicate missing data and trigger retransmission before the timer expires.

Example

Suppose a sender transmits three segments: 1, 2 and 3.

- Segment 1 reaches the receiver.
- Segment 2 is lost.
- Segment 3 arrives, but the receiver is still missing segment 2.
- TCP detects the missing data and retransmits it.
- The receiver reassembles the byte stream in order before delivering it to the application.

Key point: Error control provides TCP's reliable data delivery.

Congestion Control in TCP

Congestion control prevents excessive traffic from overloading the network.

Network congestion occurs when too much data enters the network, causing queues to grow, delays to increase and packets to be dropped.

TCP adjusts its transmission rate according to signs of network congestion.

Application Layer:

Application layer protocols are those protocols utilized at the application layer of the OSI (Open Systems Interconnection) and TCP/IP models. They facilitate communication and data sharing between software applications on various network devices. These protocols define the rules and standards that allow applications to interact and communicate quickly and effectively over a network.

Note: A critical part of this layer is the set of Application Layer Protocols, which define the rules, message formats and processes for communication between applications across different systems. These protocols standardize how data is requested, transferred and presented.

Common Application Layer Protocols in Computer Networks

1. TELNET

TELNET Provides remote login and terminal emulation services.

Allows users to access resources of a remote server as if logged in locally.

Port Number: 23

Command:

telnet [\\RemoteServer]

\\RemoteServer

:Specifies the name of the server

to which you want to connect

2. FTP

FTP stands for File Transfer Protocol. It is the protocol that actually lets us transfer files. It can facilitate this between any two machines using it. But FTP is not just a protocol but it is also a program.

FTP promotes sharing of files via remote computers with reliable and efficient data transfer.

The Port number for FTP is 20 for data and 21 for control.

Command

ftp machinename

3. TFTP

TFTP is a Simplified version of FTP, suitable for quick file transfers without authentication.

Often used for transferring boot files or firmware updates.

Port: 69

Command

tftp [options...] [host [port]] [-c command]

4. NFS

NFS Allows remote systems to mount and access files over a network as if they were local.

Used for resource centralization.

Port: 2049

Command

service nfs start

5. SMTP

Used for sending emails across networks.

Works with Mail Transfer Agents (MTA) for message forwarding.

Port: 25 (SMTP), also 587 for secure transmission.

Command

MAIL FROM:<mail@abc.com?

6. LPD

Designed for printer sharing over a network.

Receives and processes print requests.

Port: 515

Command

lpd [-d] [-l] [-D DebugOutputFile]

7. X window

Provides a graphical user interface (GUI) environment for client-server applications.

Commonly used in UNIX/Linux systems.

Port: 6000+ (increments for each server)

Command

Run xdm in runlevel 5

8. SNMP

SNMP is Used for network device management and monitoring.

Allows administrators to query devices and receive alerts.

Ports: 161 (TCP), 162 (UDP)

Command:

snmpget -mALL -v1 -cpublic snmp_agent_Ip_address sysName.0

9. DNS

Translates domain names into IP addresses (e.g., www.abc.com -> 198.105.232.4).

Essential for web browsing.

Port: 53

Command:

ipconfig /flushdns

10. DHCP

DHCP Assigns IP addresses dynamically to hosts on a network.

Provides additional configuration details like default gateway and DNS.

Ports: 67 (Server), 68 (Client)

Command:

clear ip dhcp binding {address | * }

11. HTTP/HTTPS

HTTP is used for accessing web resources, while HTTPS adds encryption for secure transactions.

Stateless and client-server based.

Ports: 80 (HTTP), 443 (HTTPS)

Command:

curl http://example.com

12. POP

POP stands for Post Office Protocol and the latest version is known as POP3 (Post Office Protocol version 3). This is a simple protocol used by User agents for message retrieval from mail servers.

POP protocol work with Port number 110.

It uses TCP for establishing connections.

Note: POP works in dual mode- Delete mode, Keep Mode. In Delete mode, it deletes the message from the mail server once they are downloaded to the local system. In Keep mode, it doesn't delete the message from the mail server and also facilitates the users to access the mails later from the mail server.

13. IRC

IRC Provides real-time text messaging for one-to-one or group communication.

Supports file and media sharing.

Port: 6667

Command:

irc [server]

14. MIME

MIME Extends email capabilities to support multimedia like images, video and programs.

Works alongside SMTP to transmit non-text data.

Not a standalone protocol, but an extension.

15. NNTP (Network News Transfer Protocol)

NNTP allows reading and posting messages to Usenet newsgroups. Supports message exchange and discussion forums.

Port: Typically TCP 119.

Use case: Online forums and discussion boards before the rise of web forums.

Command:

nntp [server]

16. LDAP (Lightweight Directory Access Protocol)

LDAP provides access and management of directory services over a network. Supports user authentication, resource lookup, and directory management.

Port: Typically TCP/UDP 389, secure version (LDAPS) uses TCP 636.

Use case: Active Directory in Windows, corporate directories.

Command:

ldap [server]

17. NTP (Network Time Protocol)

NTP synchronizes system clocks across networked devices. Supports accurate timestamps for logging and security.

Port: UDP 123.

Use case: Keeping servers and network devices in sync, e.g., for financial transactions or logging.

Command:

ntp [server]

18. MQTT (Message Queue Telemetry Transport Protocol)

MQTT provides lightweight messaging for IoT devices. Supports publish/subscribe communication between devices and brokers.

Port: Typically TCP 1883, secure version (MQTTS) uses TCP 8883.

Use case: IoT sensors, home automation, telemetry systems.

Command:

mqtt [broker] [topic]

19. SIP (Session Initiation Protocol)

SIP establishes, manages, and terminates multimedia sessions. Supports VoIP calls, video calls, and conferencing.

Ports: TCP/UDP 5060 (non-encrypted), TCP/UDP 5061 (encrypted with TLS).

Use case: VoIP services, video conferencing, and instant messaging apps.

Command:

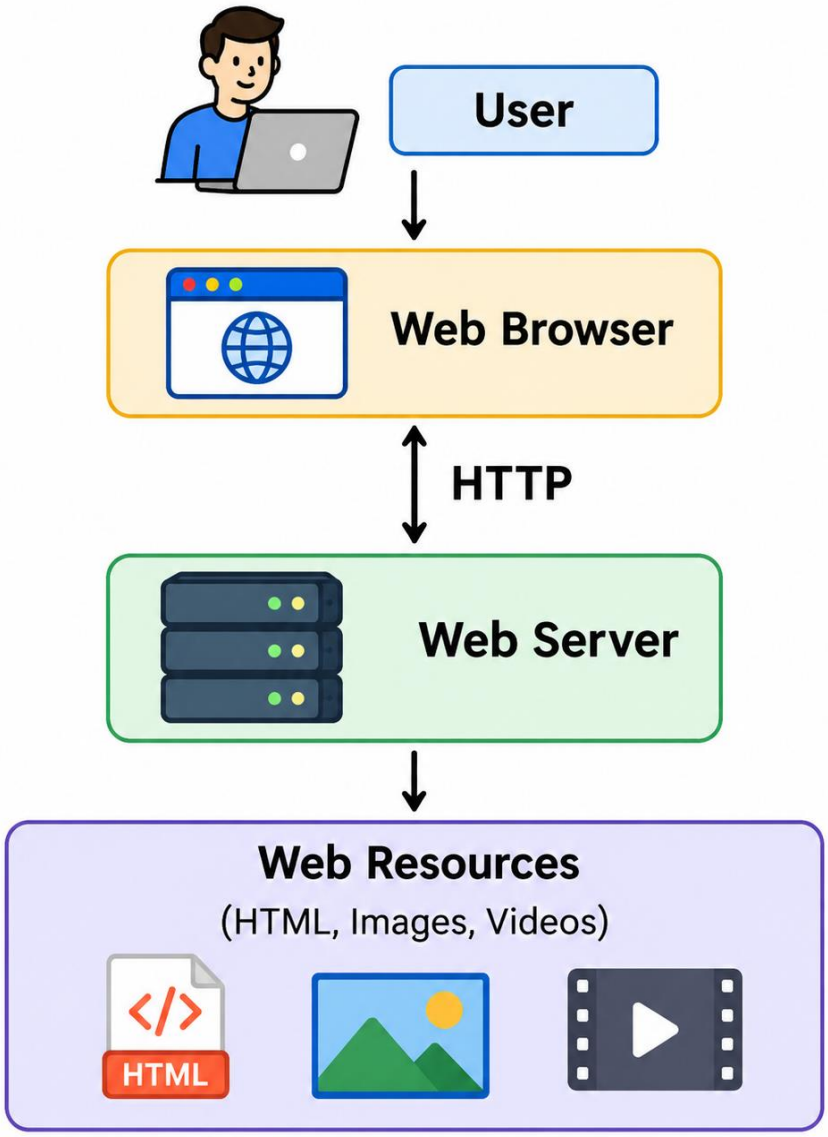
sip [server]

World Wide Web (WWW)

The World Wide Web is a distributed system of interconnected web pages and resources available through the Internet.

A user accesses web resources using a web browser such as Chrome, Firefox, or Edge.

Basic WWW Architecture



Important Components

1. Web Browser – Client application used to access websites.
2. Web Server – Stores and provides web resources.
3. URL – Identifies the location of a web resource.
4. HTTP/HTTPS – Protocol used for communication between browser and server.
5. HTML – Language used to create web pages.

Example

When a user enters:

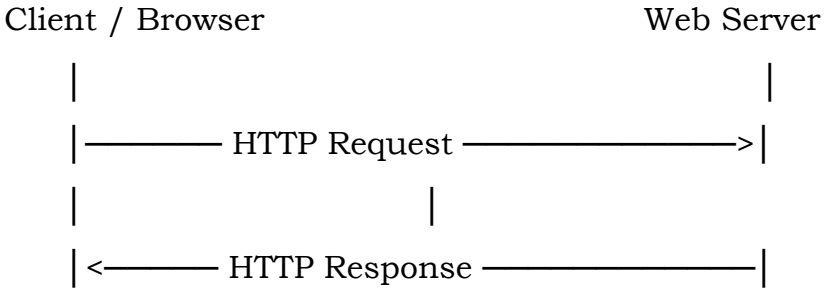
www.example.com

the browser sends a request to the web server, and the server returns the requested web page.

HTTP – HyperText Transfer Protocol

HTTP is an application-layer protocol used to transfer web pages and other resources between a client and a web server.

HTTP follows a client-server model.



HTTP Request

A request is sent by the client to obtain a resource.

Example:

GET /index.html HTTP/1.1

Host: example.com

HTTP Response

The server responds with the requested resource and a status code.

Example:

HTTP/1.1 200 OK

Content-Type: text/html

Common HTTP Methods

Method	**Purpose**
GET	Retrieve data
POST	Submit data
PUT	Update a resource
DELETE	Delete a resource
HEAD	Retrieve header information

Common HTTP Status Codes

Code	**Meaning**
200	OK / Successful
301	Moved Permanently
400	Bad Request
401	Unauthorized

Code	Meaning
403	Forbidden
404	Not Found
500	Internal Server Error

HTTP vs HTTPS

HTTPS = HTTP + Security

HTTPS uses **TLS encryption** to protect communication between the browser and server.

Electronic Mail

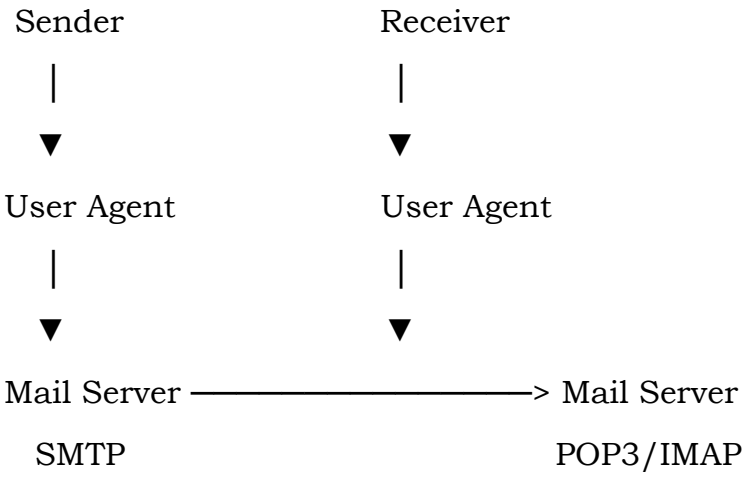
Electronic mail or **email** is an application-layer service used to exchange messages over computer networks.

Email Architecture

The major components are:

1. User Agent (UA)
2. Mail Server
3. Mail Transfer Agent (MTA)
4. Mail Access Agent

Basic Architecture



User Agent

The User Agent is the email application used by a user.

Examples:

- Gmail
- Outlook

- Thunderbird
- Apple Mail

It allows users to:

- Compose messages
- Send messages
- Receive messages
- Read messages
- Reply and forward messages

Email Protocols

Three important protocols are associated with electronic mail:

SMTP – Simple Mail Transfer Protocol

SMTP is mainly used for sending and transferring email.

Sender → Sender Mail Server → Receiver Mail Server

SMTP

SMTP generally uses:

- Port 25 – traditional SMTP
- Port 587 – message submission
- Port 465 – SMTP over implicit TLS

POP3 – Post Office Protocol Version 3

POP3 is used to retrieve email from a mail server.

A traditional POP3 model downloads messages from the server to the user's device.

Port: 110
Secure POP3: 995

IMAP – Internet Message Access Protocol

IMAP allows users to access and manage email while messages remain stored on the mail server.

Port: 143
Secure IMAP: 993

POP3 vs IMAP

POP3	IMAP
Mainly downloads emails	Keeps email synchronized with server
Suitable for single-device use	Suitable for multiple devices
Limited server-side management	Better server-side management
Simpler	More feature-rich

Web-Based Mail

Web-based mail allows users to access email using a web browser instead of a dedicated email application.

Examples include web interfaces provided by services such as Gmail and Outlook.

Architecture

User

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Web Browser

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| HTTPS

▼

Web Mail Server

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Mail Storage

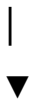
Advantages

- No email software installation required
- Accessible from any Internet-connected device
- Easy to use
- Centralized email storage
- Supports contacts, calendars, attachments, etc.

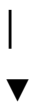
Example

- When a user opens a webmail service:

Browser → HTTPS → Web Mail Server



Mail Server



Email Data

Email Security

Email security protects messages from unauthorized access, modification, impersonation, and malicious content.

Major Email Security Issues

1. **Confidentiality** – Prevent unauthorized reading.
2. **Integrity** – Prevent unauthorized modification.
3. **Authentication** – Verify sender identity.
4. **Non-repudiation** – Prevent the sender from denying a valid communication.
5. **Protection against spam and phishing**
6. **Protection against malware**

Common Security Techniques

1. TLS

TLS encrypts email communication while it travels between systems.

2. Digital Signature

A digital signature can provide:

- Authentication
- Integrity
- Non-repudiation

3. Email Encryption

Encryption converts readable email into an unreadable form that can be restored only by an authorized recipient.

Common approaches include **S/MIME** and **OpenPGP**.

4. SPF

Sender Policy Framework (SPF) helps domain owners specify which mail servers are authorized to send mail for their domain.

5. DKIM

DomainKeys Identified Mail (DKIM) uses cryptographic signatures to help verify that a message was authorized by the sending domain and was not altered in transit.

6. DMARC

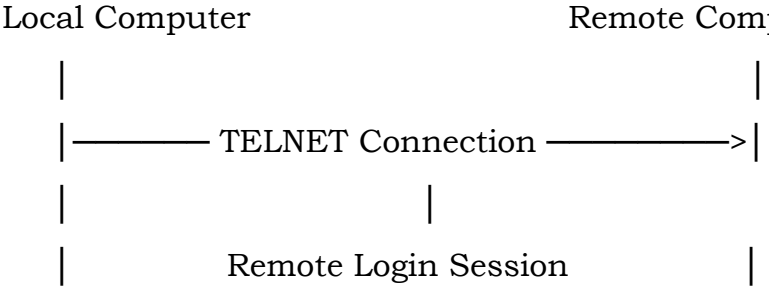
Domain-based Message Authentication, Reporting, and Conformance (DMARC) builds on SPF and DKIM to allow domain owners to specify how receiving systems should handle messages that fail authentication checks.

- **TELNET**

TELNET stands for **TELecommunication NETwork**.

It is an application-layer protocol that provides **remote terminal access** to another computer over a network.

TELNET Architecture



TELNET traditionally uses **TCP port 23**.

Working

1. User starts a TELNET client.
2. Client establishes a TCP connection with the remote machine.
3. User provides login credentials.
4. The remote system authenticates the user.
5. Commands entered by the user are executed on the remote system.

Major Problem with TELNET

TELNET sends information, including credentials, **without adequate encryption**.

Therefore, TELNET is generally unsuitable for secure remote administration.

SSH (Secure Shell) is the modern secure alternative.

- **Local vs Remote Logging**

Local Logging

In local logging, the user works directly on the computer where the application or command is executed.

User

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Local Computer

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Local Resources

Example:

A user logs into their own computer and runs commands locally.

Remote Logging

In remote logging, the user accesses another computer through a network.

User

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Local Computer

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| Network

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Remote Computer

Example:

An administrator remotely accesses a server.

Comparison

Local Logging	**Remote Logging**
Accesses local machine	Accesses remote machine
No network connection required	Network connection required
Usually faster	Depends on network
Local resources are accessed	Remote resources are accessed
Example: local terminal	Example: SSH/TELNET

Domain Name System (DNS)

DNS – Domain Name System is a distributed naming system that translates **domain names into IP addresses**.

Humans prefer names such as:

www.example.com

while computers communicate using IP addresses such as:

93.184.216.34

DNS provides the mapping between them.

Domain Name

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DNS Server

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IP Address

Why DNS is Needed?

Without DNS, users would need to remember numerical IP addresses for websites.

For example:

www.example.com

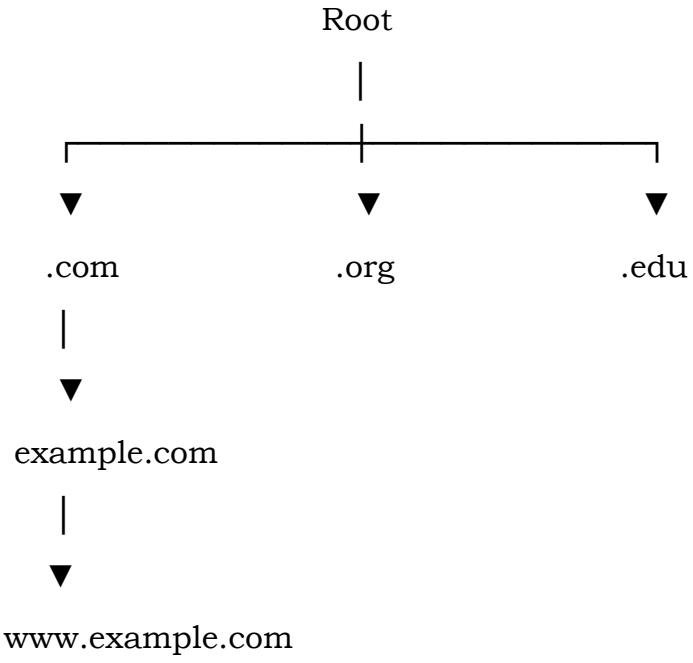
↓

DNS lookup

↓

93.184.216.34

DNS Hierarchy: DNS has a hierarchical structure.



Main Levels

1. **Root**
2. **Top-Level Domain (TLD)**
3. **Second-Level Domain**
4. **Host/Subdomain**

Examples of TLDs

- .com
- .org
- .edu
- .gov
- .in

DNS Resolution

When a user enters a website name, the DNS resolver finds the corresponding IP address.

Basic Process

User enters:

www.example.com

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DNS Resolver

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Root DNS Server

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▼

.com DNS Server

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Authoritative DNS Server

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▼

IP Address

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▼

Web Server

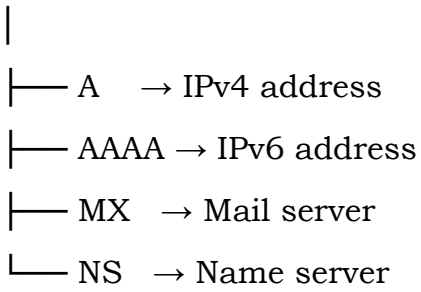
The actual process can be optimized by DNS caching, so a resolver may already have the required answer and avoid querying all levels.

Important DNS Records

Example

Record	Purpose
A	**Maps domain to IPv4 address**
AAAA	**Maps domain to IPv6 address**
MX	**Specifies mail server**
CNAME	**Alias for another domain name**
NS	**Identifies authoritative name servers**
TXT	**Stores text and policy information**
PTR	**Used for reverse DNS lookup**

example.com



Important Application Layer Protocols – Summary

Protocol	Main Function	Common Port
HTTP	**Web communication**	**80**
HTTPS	**Secure web communication**	**443**
SMTP	**Sending email**	**25 / 587**
POP3	**Receiving email**	**110**
IMAP	**Email access/synchronization**	**143**
TELNET	**Remote terminal access**	**23**
DNS	**Domain name resolution**	**53**